

SIH 2026 / HEALTHCARE AI INNOVATION

MediDecode

AI-Powered Prescription & Health Companion

"Decode. Understand. Remember. Stay Safe."

- AI Vision & Handwriting Decoder: Reads difficult clinic slips.
- Vernacular Translation: Clear instructions in 11 Indian languages.
- Food & Medication Safety: Proactive warnings on everyday diet clashes.
- Smart Reminders & Adherence: Converts Latin frequencies into clock alarms.

Presented by: Student Innovation Team | SIH 2026 Project Showcase

LIVE APP PREVIEW

Scanned Slip Decoded:

Amoxicillin 500 mg

- Timing: After Meals
- Frequency: Twice Daily (8 AM / 8 PM)
- **Confidence: 94% Verified** 

Food Safety Alert:

- Separate from dairy (milk/curd) by 2 hrs.

Regional Language:

- भोजन के बाद दिन में दो बार लें।

✓ **Zero-Guessing Clinical Safety Rule**

Healthcare Instructions Shouldn't Be Difficult to Understand

Patients and caregivers struggle daily with complex, unreadable, and foreign medical documents.



Difficult Handwritten Prescriptions

Fast cursive shorthand, smudged inks, and illegible drug names create severe misinterpretation risks for patients and pharmacies.



Complex Medical Terminology

Clinical jargon like 'Cholecystectomy', 'NPO', or 'Dyslipidemia' leaves patients completely in the dark about their actual diagnosis.



Severe Language Barriers

Over 80% of prescriptions are written in English or Latin, creating huge communication gaps for non-English speaking families across India.



Forgotten Medicine Timings

Confusing dosage frequencies (OD, BD, TID, HS) lead to skipped doses, double dosing, or prematurely stopping antibiotic courses.



Confusion About Food & Medicine

Patients unknowingly take antibiotics with milk (neutralizing absorption) or statins with grapefruit, triggering severe adverse effects.



Dense Discharge Summaries

Multi-page post-op hospital summaries overwhelm patients with clinical codes instead of simple, actionable home recovery instructions.

Meet MediDecode

An end-to-end intelligent platform bridging clinical documents and daily recovery.

STEP 01

Upload

Prescription or
Discharge Slip

STEP 02

AI Vision

Multimodal OCR &
Extraction

STEP 03

Plan

Structured Dosing &
Clock Times

STEP 04

Food Safety

Eat / Limit / Avoid
Guidance

STEP 05

Vernacular

11 Indian Native
Languages

STEP 06

Reminders

Smart Alarms &
Dashboard



Strict Healthcare Safety

Never invents or hallucinates medicine names. Unclear handwriting is flagged with confidence scores and user verification alerts.



Layperson Simplification

'Explain this to me' feature translates surgical notes and clinical terms into 2-sentence conversational language.

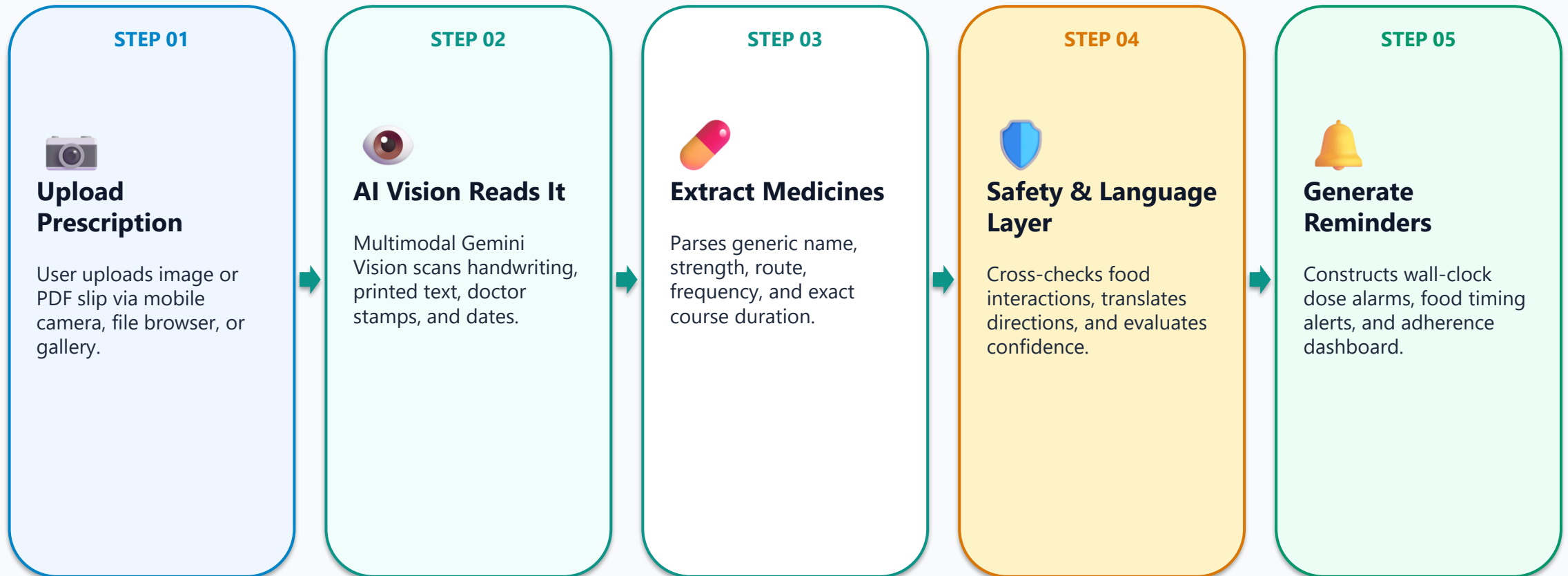


Informational Companion

Clearly reminds patients to verify important medication decisions with licensed doctors and pharmacists.

How It Works: 5-Step Clinical Intelligence Flow

From an unreadable paper slip to a personalized daily recovery companion.



PRODUCT LIFECYCLE: UPLOAD → DECODE → UNDERSTAND → CHECK → REMIND → TRACK

Core Features of MediDecode

Engineered to deliver comprehensive clinical clarity and high user adoption.

AI Prescription Scanner

Scans clinic slips, doctor notes, and stamped slips with specialized vision prompts.

Medicine Decoder

Standardizes brand names into International Nonproprietary Names (INN) with strength & route.

Smart Medication Schedule

Converts Latin shorthand (OD, BD, TID) into concrete clock alarms (8:00 AM, 8:00 PM).

Medicine Reminders

Generates web push notifications and audio alerts so patients never skip a crucial dose.

Food & Medication Guidance

Proactive dietary engine highlights foods to eat, limit, or avoid based on active medicines.

Precaution & Safety Alerts

Color-coded confidence scoring (Green/Yellow/Red) flags ambiguous cursive for pharmacist review.

Regional Language Translation

Translates instructions into 11 Indian languages while preserving standardized medicine names.

Discharge Summary Simplification

'Explain this to me' feature translates complex surgical and hospital discharge notes.

Food & Medication Safety Engine

A vital safety layer preventing common adverse diet-drug interactions.

FLOW: [Medicine Prescribed] → [Clinical Interaction Engine] → [Eat / Limit / Avoid Analysis] → [Safety Alert]

REAL-WORLD CLINICAL EXAMPLE: Ciprofloxacin 500 mg (Antibiotic) | Source: British National Formulary (BNF)

EAT (SUITABLE)

Foods that do not interfere:

- Plain cooked rice, steamed vegetables
- Water, clear soups, herbal tea
- Whole grain bread & lean proteins
- High hydration promotes safe drug clearance

LIMIT / TIME (CAUTION)

Requires strict time separation:

- Milk, Curd, Yogurt, Cheese, Paneer
- Calcium binds to antibiotic molecules
- Reduces drug absorption by up to 70%
- SAFE RULE: Separate dairy by 2 to 4 hours

AVOID / CHECK (HIGH ALERT)

Severe risks requiring attention:

- Calcium & Iron fortified supplements
- Alcohol: Triggers severe stomach distress
- Antacids containing aluminum/magnesium
- Always verify with prescribing doctor

 **CLINICAL SAFETY NOTICE:** Always verify medication-specific food restrictions with a qualified doctor or pharmacist.

Smart Medication Reminders & Daily UI

A realistic, distraction-free smartphone experience that keeps patients on track.

09:41 AM ●●● Wi-Fi 🔋 98%

Good Morning, Rahul 🙌

Today's Schedule: 2 of 3 Doses Taken

🕒 **09:00 AM — Amoxicillin 500 mg**

• 1 tablet | After breakfast

• **STATUS: TAKEN ✓ (09:05 AM)**

🕒 **02:00 PM — Pantoprazole 40 mg**

• 1 tablet | Before lunch

• **STATUS: NEXT DOSE ⌚ (In 45 mins)**

🕒 **09:00 PM — Amoxicillin 500 mg**

• 1 tablet | After dinner

• **STATUS: UPCOMING 📅**

🟢 Taken ✓ (1-Tap Checkoff)

Patients easily tap their doses. Confetti animations and adherence scores celebrate consistency.

🕒 Next Dose (Smart Countdown)

Highlights the immediate upcoming medicine with food instructions (e.g. 'Before lunch').

🌙 Upcoming (Daily Overview)

Shows evening and bedtime doses so patients and family caregivers can plan meals in advance.

🔴 Missed / Overdue Tracking

If a dose is delayed, gentle reminder notifications prompt the patient and notify family members.

Regional Language Support

Bridging the linguistic divide to make healthcare universally accessible.

"Healthcare information becomes truly useful only when patients can understand it in their own language."

GB English (Original)

Standard Medical Direction

Amoxicillin 500 mg

Take 1 tablet twice daily after meals with water.

IN Hindi (हिन्दी)

National Vernacular

Amoxicillin 500 mg

भोजन के बाद पानी के साथ दिन में दो बार 1 गोली लें।

IN Kannada (ಕನ್ನಡ)

Regional State Language

Amoxicillin 500 mg

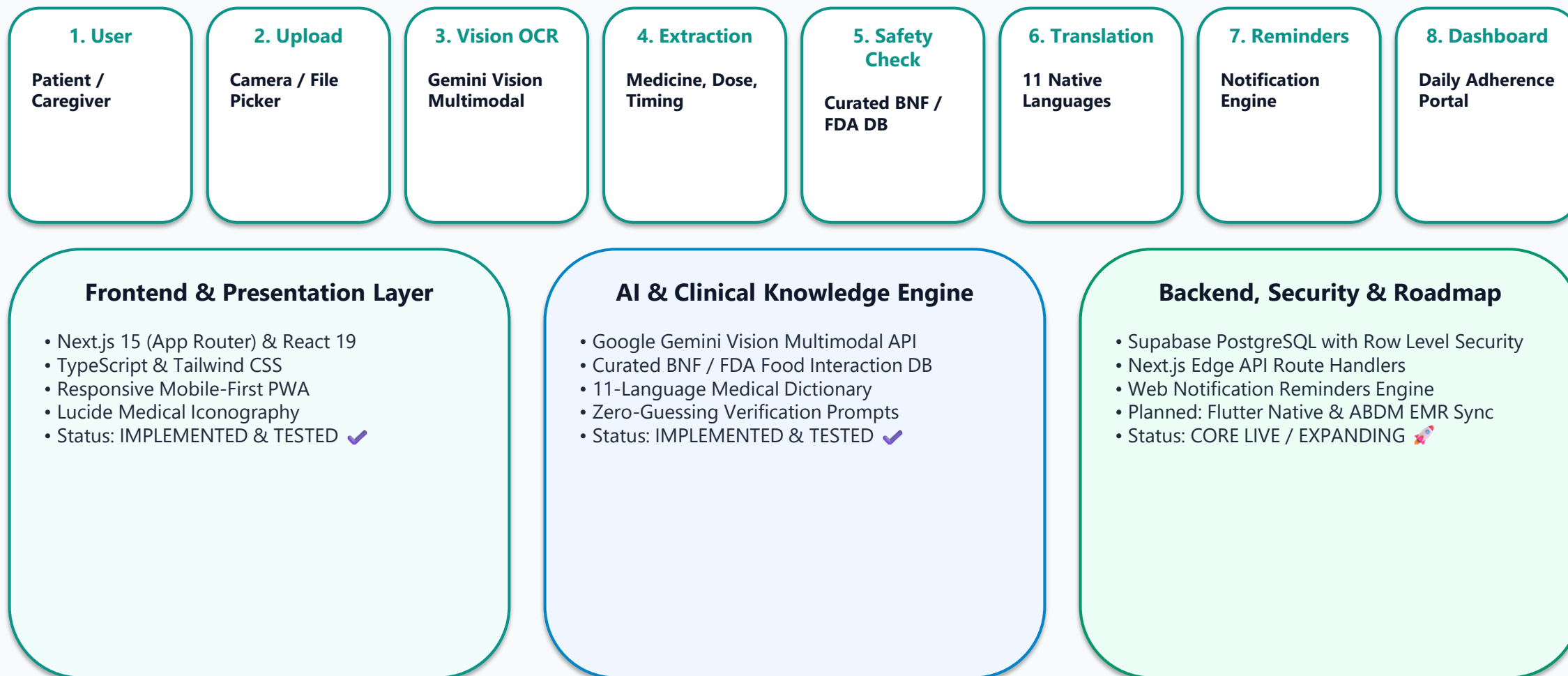
ಊಟದ ನಂತರ ದಿನಕ್ಕೆ ಎರಡು ಬಾರಿ ನೀರಿನೊಂದಿಗೆ 1 ಮಾತ್ರೆ ತೆಗೆದುಕೊಳ್ಳಿ.

KEY CLINICAL SAFETY RULE IN TRANSLATION:

Medicine names (e.g. 'Amoxicillin 500 mg') are preserved in Roman script to prevent dangerous drug confusion at local pharmacy counters, while frequencies and food timings translate seamlessly into the patient's native script.

Technology Architecture & Data Pipeline

Decoupled full-stack architecture built for high reliability and patient privacy.



Why MediDecode? Traditional Approach vs MediDecode

Transforming passive, confusing paper slips into an active, verified digital recovery companion.

✗ TRADITIONAL HEALTHCARE APPROACH

- Difficult Cursive Handwriting: Misread drug names and wrong dosages leading to medical emergencies.
- Complex Latin Abbreviations: Terms like 'BD', 'TID', 'HS' confuse patients about exact clock timings.
- English-Only Language Barrier: Non-English speakers cannot understand their own prescribed medications.
- Manual Medicine Tracking: Patients rely on memory, leading to missed doses or early course termination.
- Zero Dietary Awareness: Patients take medicines with calcium or citrus without knowing it blocks absorption.
- Dense Discharge Papers: Multi-page technical hospital documents that patients cannot interpret at home.

✓ THE MEDIDECODE PLATFORM

- AI-Assisted Multimodal OCR: Reads printed and cursive text with confidence badges and zero-guessing rules.
- Plain-Language Clock Timelines: Converts 'BD after food' into concrete 8:00 AM and 8:00 PM alarm slots.
- 11 Regional Indian Languages: Instructions in Hindi, Bengali, Kannada, Tamil, etc., while preserving INN drug names.
- Smart Automated Reminders: Web push notifications and 1-tap taken checkoff drive daily treatment adherence.
- Curated Food & Safety Engine: Proactive alerts on everyday foods to Eat, Limit, or Avoid based on verified BNF data.
- 'Explain This to Me' Summaries: Translates complicated post-op hospital records into 2-sentence actionable advice.

Real-World Impact & Beneficiaries

Empowering patients, families, and healthcare systems across India.

“MediDecode helps patients understand and follow healthcare instructions more confidently.”

 **Elderly Patients:** Large fonts, simple alarm clocks, and zero confusing medical jargon make daily medicine management effortless.

 **Families & Caregivers:** Enables children and relatives to monitor their elderly parents' doses and recovery schedules remotely.

 **Polypharmacy Patients:** Chronic patients managing 5+ daily medications get an organized, non-conflicting master timetable.

IN Regional-Language Users: Removes English literacy barriers, democratizing healthcare accessibility across Tier 2, 3, and rural India.

 **Hospital Discharge Patients:** Clear post-surgery wound care and dietary guidelines prevent avoidable emergency readmissions.

Future Scope & Vision for MediDecode

Scaling MediDecode into a comprehensive universal health companion.



FUTURE INNOVATION ROADMAP

- Voice Assistant & Audio Readout: Natural voice instructions in 11 languages for illiterate or visually impaired patients.
- 22 Official Indian Languages: Expanding vernacular dialect coverage across all Indian states.
- Doctor & Pharmacist Verification Portal: Enables local chemists to verify flagged cursive medicines in 1-click.
- Dedicated Caregiver Companion App: Real-time alerts to family members if an elderly patient misses an insulin or BP dose.
- Wearable & Smartwatch Integration: Haptic wrist buzzers for discrete, timely medication alerts on smart bands.
- ABDM & EMR Integration: Seamless sync with Ayushman Bharat Digital Mission and electronic health records.



MediDecode

"Decode. Understand. Remember. Stay Safe."

Thank you respected judges and professors for your time and guidance.

**We are now open for
Questions & Answers / Live Demo!**